

PROJECT REPORT

EMPLOYEE MANAGEMENT SYSTEM

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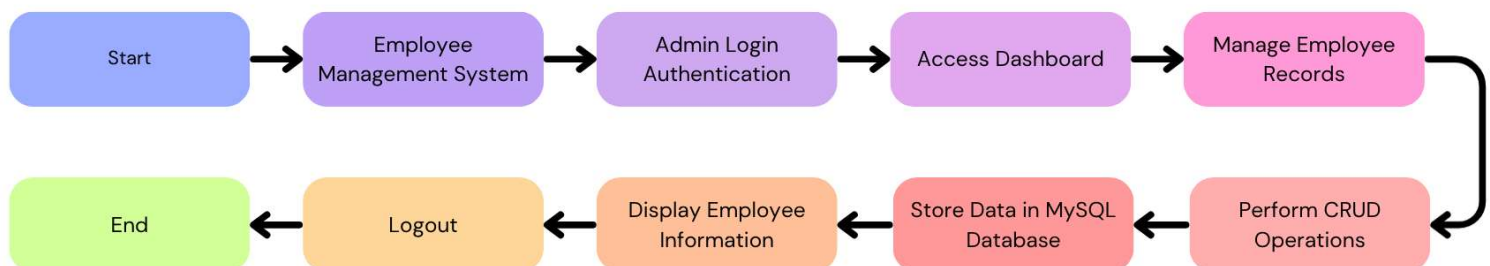
Introduction

The Employee Management System is a web-based application developed using PHP and MySQL as part of the Full Stack Web Development Internship at Prodigy InfoTech. The main purpose of this project is to manage employee records efficiently through CRUD (Create, Read, Update, Delete) operations.

This system allows the admin to add new employees, view employee records, update employee details, and delete employee data from the database. The project also includes login authentication and form validation to improve security and usability.

The frontend of the project was developed using HTML, CSS, and JavaScript, while PHP was used for backend development and MySQL was used for database management.

This project helped in understanding frontend-backend integration, database connectivity, authentication systems, and practical implementation of full stack web development concepts.



Project Objective

The main objective of this project is to develop a simple and efficient Employee Management System using PHP and MySQL. The system is designed to manage employee records through CRUD (Create, Read, Update, Delete) operations.

The project aims to provide a user-friendly interface where the admin can securely log in and perform employee management tasks such as adding, viewing, updating, and deleting employee information.

Objectives of the project include:

- To understand frontend and backend integration
- To learn database connectivity using MySQL
- To perform CRUD operations using PHP
- To implement authentication and validation
- To improve practical knowledge of full stack web development
- To gain hands-on experience in web application development

Technologies Used

The following technologies were used in the development of the Employee Management System project:

Technology	Purpose
HTML	Used to create the structure of web pages
CSS	Used for styling and designing the user interface
JavaScript	Used for form validation and basic interactivity
PHP	Used for backend development and server-side scripting
MySQL	Used for storing employee and admin data
XAMPP	Used as a local server environment
phpMyAdmin	Used to manage the MySQL database
VS Code	Used as the code editor for project development

These technologies helped in building a complete full stack web application with database connectivity and CRUD functionality.

Project Features

The Employee Management System includes the following features:

- Admin Login Authentication
- Secure Login System
- Add New Employee Records
- View Employee Details
- Update Employee Information
- Delete Employee Records
- Form Validation
- Logout Functionality
- Database Connectivity using MySQL
- User-Friendly Interface

The system allows the admin to efficiently manage employee data through a simple and interactive web interface.

System Requirements

The following hardware and software requirements are needed to run the Employee Management System project successfully.

Hardware Requirements

- Minimum 4 GB RAM
- Intel i3 Processor or higher
- 500 MB Free Disk Space
- Keyboard and Mouse

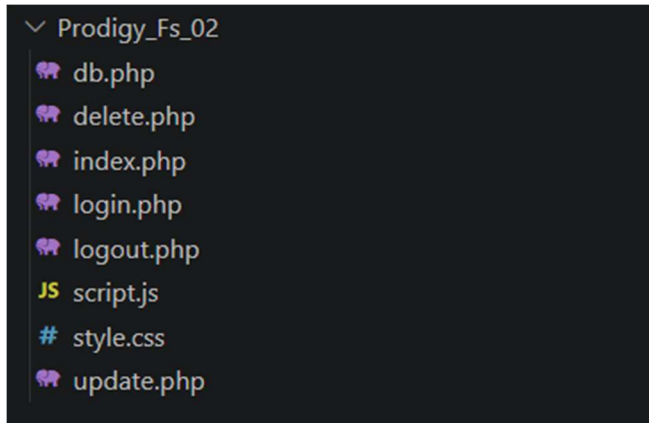
Software Requirements

- Windows Operating System
- XAMPP Server
- PHP
- MySQL Database
- phpMyAdmin
- Visual Studio Code (VS Code)
- Web Browser (Google Chrome / Microsoft Edge)

These requirements provide the necessary environment for developing and running the project smoothly.

Project Structure

The Employee Management System project is organized into multiple files, where each file performs a specific function in the application.



Description of Files

File Name	Purpose
index.php	Main dashboard page used for employee management
login.php	Handles admin login authentication
logout.php	Used to logout from the system
update.php	Updates employee details
delete.php	Deletes employee records
db.php	Establishes database connection
Style.css	Contains styling and design of the project
script.js	Used for JavaScript functionality
README.md	Contains project documentation

This structure helps in organizing the project files properly and improves maintainability of the application.

Database Setup

The Employee Management System uses MySQL as the database for storing employee and admin information. The database was created and managed using phpMyAdmin in XAMPP.

Database Name

employee_db

Employees Table

The employees table stores all employee details such as name, email, department, and salary.

```
CREATE TABLE employees (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100),  
  email VARCHAR(100),  
  department VARCHAR(100),  
  salary INT  
);
```

Admin Table

The admin table is used for login authentication.

```
CREATE TABLE admin (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  username VARCHAR(100),  
  password VARCHAR(100)  
);
```

Insert Admin Login Credentials

```
INSERT INTO admin(username, password)  
VALUES('admin', '12345');
```

Database Connectivity

The database connection was established using PHP and MySQL through the db.php file.

The following details were used:

Parameter	Value
Host	localhost
Username	root
Password	(empty)
Database Name	employee_db

This setup enables the project to perform CRUD operations and authentication successfully.

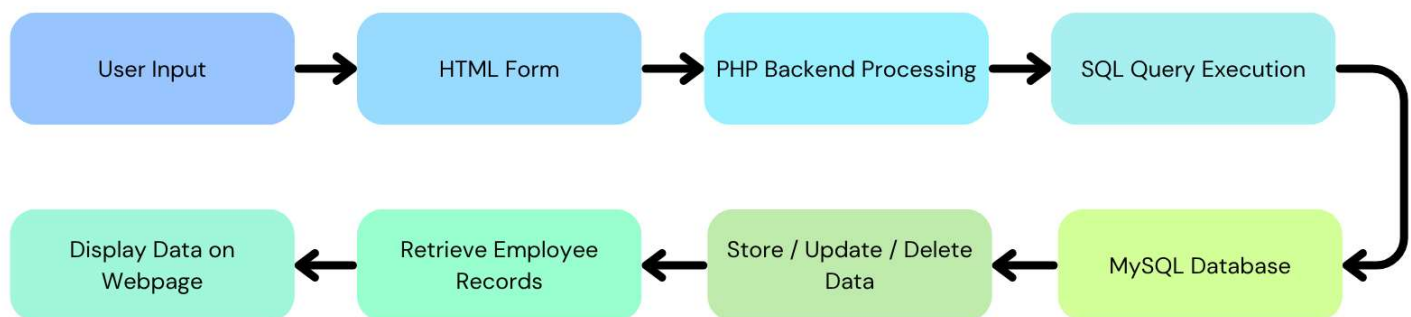


Figure : Database Flowchart

How to Run the Project

The following steps are used to run the Employee Management System project successfully.

Step 1: Install XAMPP

Download and install XAMPP on the system.

Step 2: Start Apache and MySQL

Open the XAMPP Control Panel and start:

- Apache
- MySQL

Step 3: Move Project Folder

Copy the project folder Prodigy_FS_02 into:

<C:\xampp\htdocs>

Step 4: Open phpMyAdmin

Open the following URL in the browser:

<localhost/phpmyadmin>

Create the required database and tables.

Step 5: Run the Project

Open the browser and enter:

localhost/Prodigy_FS_02/login.php

Step 6: Login to the System

Use the following credentials:

Username Password

admin 12345

After successful login, the Employee Management System dashboard will open successfully.

Working of the Project

The Employee Management System works through a simple and user-friendly process. The application starts with an admin login page where the admin enters valid login credentials.

After successful authentication, the admin is redirected to the main dashboard of the system. The dashboard contains a form to add employee details such as name, email, department, and salary.

When the admin submits the form, the employee data is stored in the MySQL database using PHP and SQL queries. All employee records are displayed in a table format on the same page.

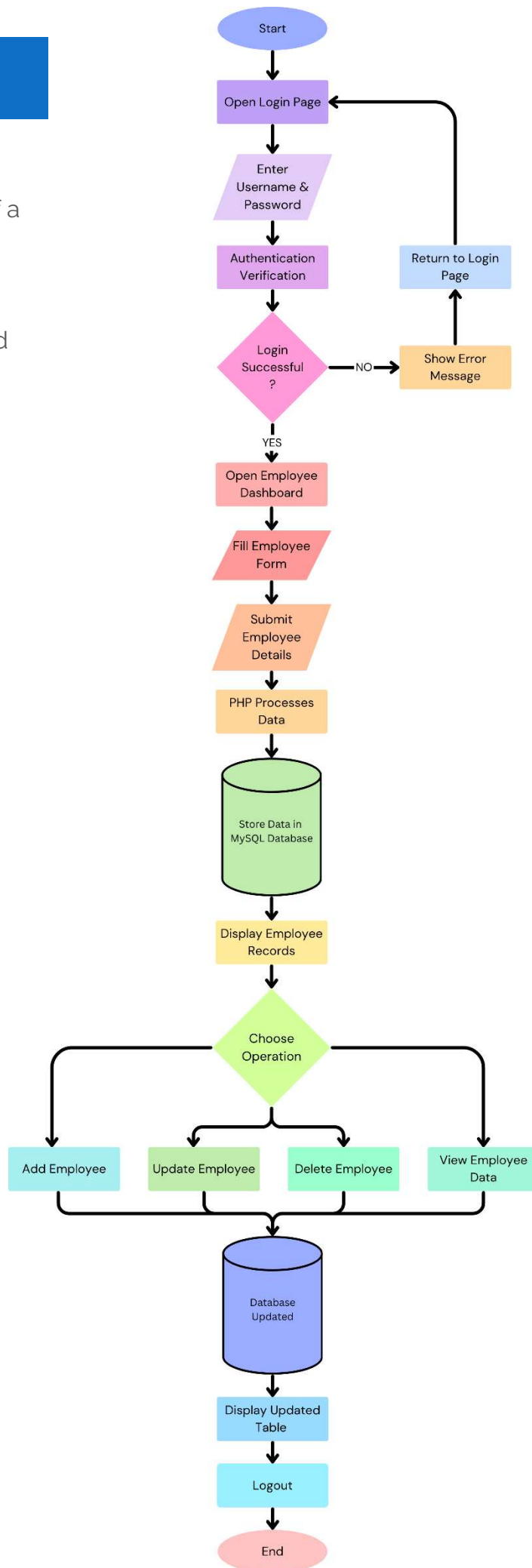
The admin can perform the following operations:

- Add new employee records
- View employee information
- Update employee details
- Delete employee records

The project also includes form validation to prevent invalid data submission. A logout option is provided to return the user to the login page.

The backend logic is implemented using PHP, while MySQL is used for database management and storage.

The project demonstrates the complete workflow of a basic full-stack authentication system using frontend technologies, backend development, session management, and database integration.



Authentication Flow

The Employee Management System includes a basic authentication mechanism to provide secure access to the admin dashboard.

The authentication process works as follows:

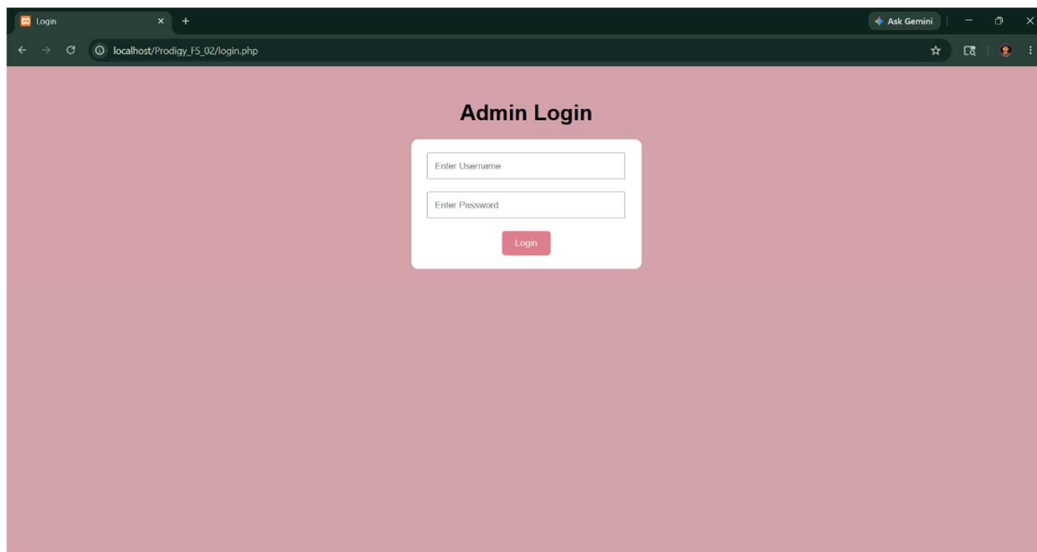
1. The admin opens the login page of the application.
2. The admin enters the username and password.
3. PHP receives the login credentials using the POST method.
4. The entered credentials are checked against the admin table in the MySQL database.
5. If the username and password are correct:
 - The admin is redirected to the Employee Management dashboard.
6. If the credentials are incorrect:
 - An error message is displayed saying "Invalid Username or Password".
7. After completing operations, the admin can logout using the Logout button.

The authentication system helps in restricting unauthorized access to the employee records and improves the security of the application.

Screenshots

The following screenshots show the working of the Employee Management System project.

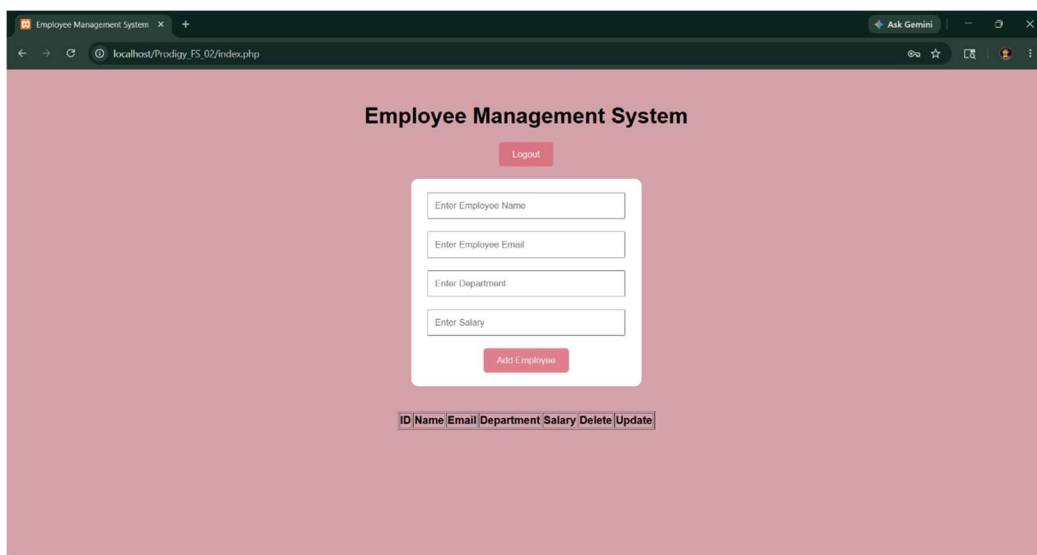
Figure 1: Login Page



Description:

This page allows the admin to enter login credentials and access the system.

Figure 2: Dashboard Page



Description:

This is the main dashboard where employee management operations are performed.

Figure 3: Add Employee Form

The screenshot shows a web browser window with the title "Employee Management System". The URL is "localhost/Prodigy_PS_02/index.php". The page has a pink background. At the top, there is a "Logout" button. Below it is a form with four input fields: "Enter Employee Name", "Enter Employee Email", "Enter Department", and "Enter Salary". At the bottom of the form is an "Add Employee" button. Below the form is a table with the following data:

ID	Name	Email	Department	Salary	Delete	Update
18	Aman	aman@gmail.com	IT	25000	Delete	Update
20	Mohit	mohit@gmail.com	HR	50000	Delete	Update

Description:

The admin can add employee details such as name, email, department, and salary. All employee records are displayed in tabular format with update and delete options.

Figure 5: Update Employee Page

The screenshot shows a web browser window with the title "Update Employee". The URL is "localhost/Prodigy_PS_02/update.php?id=18". The page has a pink background. At the top, there is a "Update Employee" button. Below it is a form with four input fields: "Aman", "aman@gmail.com", "IT", and "25000". At the bottom of the form is an "Update Employee" button.

Description:

This page allows the admin to modify employee details.

Figure 7: Database Table in phpMyAdmin

				id	name	email	department	salary
☐	 Edit	 Copy	 Delete	18	Aman	aman@gmail.com	IT	25000
☐	 Edit	 Copy	 Delete	20	Mohit	mohit@gmail.com	HR	50000
☐ Check all				With selected:				
				 Edit				
				 Copy				
				 Delete				
				 Export				

Description:

Employee records stored in the MySQL database are shown in phpMyAdmin.

Challenges Faced

During the development of the Employee Management System project, several challenges were faced while implementing different functionalities.

Some of the major challenges faced during the project are:

- Understanding the connection between PHP and MySQL
- Fixing XAMPP MySQL startup issues
- Handling CRUD operations correctly
- Debugging PHP syntax errors
- Managing form validation
- Connecting frontend and backend components
- Implementing login authentication
- Understanding SQL queries and database operations

These challenges helped in improving problem-solving skills and provided practical experience in full stack web development.

Learning outcome

Through the development of the Employee Management System project, several important concepts and practical skills were learned.

The major learning outcomes of this project are:

- Understanding CRUD operations using PHP and MySQL
- Learning database connectivity and SQL queries
- Gaining knowledge of frontend and backend integration
- Implementing login authentication systems
- Applying form validation techniques
- Improving debugging and problem-solving skills
- Understanding project structure and file organization
- Gaining practical experience in full stack web development

This project helped in strengthening both technical knowledge and practical implementation skills related to web application development.

Future Improvements

The Employee Management System can be further improved by adding more advanced features and improving the overall functionality of the application.

Some future improvements that can be implemented are:

- Adding employee search functionality
- Implementing password encryption for better security
- Adding session management for secure authentication
- Making the project fully responsive for mobile devices
- Adding employee profile pictures
- Implementing pagination for large employee records
- Adding sorting and filtering options
- Improving the user interface with advanced CSS frameworks
- Deploying the project online for public access

These improvements can make the system more secure, scalable, and user-friendly.

Conclusion

The Employee Management System project was successfully developed using PHP and MySQL as part of the Full Stack Web Development Internship at Prodigy InfoTech.

The project successfully implemented CRUD operations including adding, viewing, updating, and deleting employee records. Authentication and validation features were also implemented to improve security and usability.

Through this project, practical knowledge of frontend and backend integration, database connectivity, and web application development was gained. The project also helped in improving problem-solving, debugging, and programming skills.

Overall, this project provided valuable hands-on experience in full stack web development and enhanced understanding of real-world web application development concepts.

Acknowledgements

I would like to express my sincere gratitude to Prodigy InfoTech for providing me with the opportunity to complete this Full Stack Web Development Internship and enhance my practical knowledge in web development.

I would also like to thank the various platforms, tools, technologies, and learning resources that supported me throughout the development of the Employee Management System project.

The following resources played an important role in understanding concepts, solving problems, and completing the project successfully:

- HTML Official Documentation — <https://html.spec.whatwg.org/>
- CSS Official Documentation — <https://www.w3.org/Style/CSS/>
- JavaScript Documentation (MDN) — <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
- PHP Official Documentation — <https://www.php.net/docs.php>
- MySQL Official Website — <https://www.mysql.com/>
- XAMPP Official Website — <https://www.apachefriends.org/>
- phpMyAdmin Documentation — <https://www.phpmyadmin.net/>
- Visual Studio Code — <https://code.visualstudio.com/>
- Git Official Website — <https://git-scm.com/>
- GitHub Platform — <https://github.com/>
- YouTube Learning Platform — <https://www.youtube.com/>
- Gmail Communication Service — <https://mail.google.com/>

I sincerely appreciate all these resources for their valuable documentation, tutorials, tools, and guidance that contributed to the successful completion of this project. Finally, I would like to thank everyone who directly or indirectly supported me during this internship and helped me improve my technical and problem-solving skills.

THANKYOU